

Syllabus of Botany (Assistant Professor) for Entrance Examination

(Public Service Commission)

Unit-1: Microbiology, Mycology and Plant Pathology

Viruses: General Characteristics, ultrastructure and replication of TMV, CaMV and TYMV; Isolation, purification and chemical nature of TMV; modes of transmission and prevention; common viral diseases of plants; structural characteristics of bacteriophages (λ , T4), viroids and prions.

Bacteria: General account of Eubacteria (size, shape and arrangement) and Archaeobacteria (Methanogenic, halophilic and thermo-acidophilic); classification (Haeckel's three kingdom concept and Three Domain Concept of Carl Woese); Ultrastructure of cell wall and cell membrane, modes of reproduction in bacteria; salient features and importance of cyanobacteria; role of microbes in nanotechnology and bioremediation.

Fungi: Classification of fungi (Alexopolous & Mims 1979); ultrastructure and composition of fungal cell wall; nutritional types (biotrophic and symbiotic) of fungi; Heterothallism, heterokaryosis and parasexuality.

Plant Pathology: Disease symptoms, Etiology and control of important fungal diseases of crop plants (Brown spot of rice, Apple scab, Paddy blast, Powdery Mildew of cucurbits, White blister of crucifers, Downy mildew of vegetables, Tikka Disease of groundnut); Diseases caused by and disease symptoms of viroids (potato spindle tuber) and phytoplasma (Aster yellows).

Unit-2: Diversity and Evolution of Lower Plants

Algae: Range of thallus structure in algae; evolutionary trends; modes of reproduction (vegetative, asexual and sexual); major patterns of life cycles (mono-, di- and trigenetic types); salient features of Cyanophyta, Chlorophyta, Bacillariophyta, Chrysophyta, Dinophyta, Euglenophyta, Cryptophyta, Xanthophyta, Phaeophyta and Rhodophyta; modern trends in classification of algae; structure and diversity of chloroplasts, pyrenoids and reserve storage products; algal blooms, biofertilizers and biofuels.

Bryophyta: Recent trends in classification of bryophytes; origin, evolution and alternation of generation in bryophytes; morphology, anatomy and reproduction in Marchantiales, Metzgeriales, Jungermanniales, Anthocerotales, Sphaerocarpaceae, Polytrichales, Funariales and Sphagnales; Bryophytes as indicators of environmental pollution. Symbiotic associations, biologically active compounds of bryophytes.

Pteridophytes: Classification of Pteridophytes (Sporne 1975); Origin and evolution of Pteridophytes; evolution of stelar types, heterospory and seed habit; Morphology, anatomy and reproduction of fossil (*Rhynia*, *Trimerophyton*, *Lepidodendron*, *Sphenophyllum*) and living (*Psilotum*, *Lycopodium*, *Selaginella*, *Isoetes*, *Equisetum*, *Osmunda*, *Schizaea*, *Pteris*, *Ophioglossum*, *Marrattia*, *Dryopteris*, *Ceratopteris*, *Platyzoma*, *Asplenium*, *Acrostichum*, *Marsilea* and *Salvinia*) pteridophytes; role as biofertilizers; economic and ecological importance of pteridophytes.

Gymnosperms: Distribution of living gymnosperms in J&K; origin, evolution and classification (Sporne 1965) of gymnosperms; morphology, anatomy and reproduction in *Cycadeoidea*, *Cordaites*, *Cycas*, *Ginkgo*, *Pinus*, *Ephedra*, *Welwitschia* and *Gnetum*.

Unit-3: Taxonomy and Systematics of Angiosperms

Classification: Systems of Classification-Artificial (Carl Linnaeus), Natural (Bentham and Hooker) and Phylogenetic (Takhtajan) systems; concept and principles of phenetics and cladistics.

Taxonomic Tools: Taxonomic categories and hierarchy; species concept; keys, field and herbarium techniques; methods of plant identification (literature, e-floras, manuals, icons, journals, etc.).

Nomenclature: ICN, principles of plant nomenclature, names of taxa, type method, author citation, principles of priority; Basionyms, Synonyms, Homonyms, Autonyms and Tautonyms.

Family Description: Taxonomic description of Ranunculaceae, Fabaceae, Malvaceae, Asteraceae, Solanaceae, Poaceae, Liliaceae. Orchidaceae, Zingiberaceae.

Concept and utility of APG-IV classification; Parsimony, Maximum likelihood, and DNA barcoding in phylogenetic analysis.

Unit-4: Cytology, Genetics and Cytogenetics

Chromatin Organization and Gene Structure: Organization of chromosomes–nucleosome, centromere and telomere; euchromatin and heterochromatin; Operon, unique and repetitive DNA, Concept of gene, allele, multiple alleles, pseudoalleles, interrupted genes, gene families; complementation tests; mitochondrial and chloroplast genomes.

Genetic Recombination and Gene Mapping: Linkage and independent assortment, crossing over, linkage maps, tetrad analysis, mapping with molecular markers, mapping by using somatic cell hybrids, development of mapping population in plants; Genetic transformation, conjugation, transduction and sex-duction, mapping genes by interrupted mating.

Mutations: Spontaneous and induced mutations; Types, causes and detection; mutant types – lethal, conditional, biochemical, loss of function, gain of function; germinal versus somatic mutants; insertional mutagenesis; transposable elements in eukaryotes (Ac-Ds, P and Ty elements) and prokaryotes (IS, Tn3 and Tn10).

Structural and Numerical Alterations of Chromosomes: Deletion, duplication, inversion, translocation; Euploidy (auto and allo-polyploidy) and aneuploidy (monosomics and trisomics) and their genetic implications.

Unit-5: Cell and Molecular Biology

Plant Cell Envelope, Structure and Function: Structure, function and biogenesis of primary and secondary cell wall in plants; plasma membrane–chemical composition, organization of various components, fluid-mosaic model, lipid rafts; plasma membrane and cellular details of transport-carriers, pumps and channels; mechanism of sorting and regulation of intracellular transport, electrical properties of membranes; a general concept of cell division (Mitosis and meiosis) and cell cycle regulation.

Structural Organization and Function of Intracellular Organelles: Structure and functions of Nucleus, Mitochondria, Golgi bodies, Lysosomes, Endoplasmic reticulum, Peroxisomes, Plastids, Vacuoles; structure & function of Cytoskeleton and its role in motility.

DNA Structure, Replication and Repair: DNA structure–A, B, Z Types; mechanism of replication in prokaryotes and eukaryotes, enzymology and fidelity of replication; DNA damage and repair mechanisms.

RNA Synthesis and Processing: Transcription factors, machinery and mechanism in prokaryotes and eukaryotes; formation of initiation complex, transcription activators and repressors, RNA capping, polyadenylation, splicing and editing; structure and function of different types of RNA, mRNA transport.

Protein Synthesis and Processing: Ribosome, charging of tRNA, Mechanism of translation in prokaryotes and eukaryotes, factors involved there in; genetic code, translational proof-reading, translational inhibitors, post-translational modification of proteins.

Regulation of Gene Expression: Control at transcription and translation level in prokaryotes (lac and trp operons) and eukaryotes (hormonal control and methylation).

Unit-6: Physiology and Metabolism

Biomolecules, Enzymes and Bioenergetics: Composition, structure and function of biomolecules (carbohydrates, lipids, proteins, nucleic acids and vitamins); Enzymes and enzyme kinetics, enzyme regulation, mechanism of enzyme catalysis, isozymes; Glycolysis, oxidative phosphorylation.

Photosynthesis and Respiration: Light harvesting complexes; mechanisms of electron transport; photo-protective mechanisms; CO₂ fixation- C₃, C₄ and CAM pathways; Citric acid cycle; plant mitochondrial electron transport and ATP synthesis; alternate oxidase; photorespiration.

Nitrogen Metabolism and Plant Hormones: Nitrate and ammonium assimilation; amino acid biosynthesis; Plant Hormones-Biosynthesis, storage, breakdown and transport; physiological effects and mechanisms of action.

Transport, Photobiology and Stress Physiology: Uptake, transport and translocation of water, ions, solutes and macromolecules from soil, through cells, across membranes, through xylem and phloem; transpiration; mechanisms of loading and unloading of photo-assimilates; Structure, function and mechanisms of action of phytochromes, cryptochromes; stomatal movement; photoperiodism and biological clocks; Responses of plants to biotic (pathogen and insects) and abiotic (water, temperature and salt) stresses.

Unit-7: Plant Anatomy, Reproduction and Economic Botany

Plant Tissues & Organs: Primary structure and basic vasculature of Root, Leaf and Shoot: Origin and diversity of epidermis, stomata and trichomes; Meristems-types, composition and structure; function of RAM and SAM; organization and function of vascular cambium, Structure and function of primary and secondary Xylem and Phloem, Anatomy of root, stem and leaf in monocots and dicots.

Plant Reproduction: Types of reproduction, transition to flowering (Autonomous, photoperiodic and vernalization pathways); flower development, microsporogenesis and micro-gametogenesis; megasporogenesis and mega-gametogenesis; pollination types and syndromes, pollen-pistil interaction, male sterility and self-incompatibility, fruit and seed dispersal mechanisms; Embryo & Endosperm development (brief account).

Plant Resource Utilization: Origin of Crop Plants (primary and secondary centres); gene pool and germplasm concept; Origin and uses of Food (maize, buckwheat), Fibre (cotton, jute), Spices (saffron, ginger), Oil (groundnut, mustard), Sugar (beetroot), beverage (tea) yielding plants, bamboos and rubber.

Plant Resource Conservation: Biodiversity extinction, types and causes; IUCN categories of plants; objectives and principles of conservation and resource management – In-situ and ex-situ conservation strategies; Legislation and enactments on conservation; concept of IPR.

Unit-8: Ecology and Environment

Environment, Habitat and Niche: Physical environment; biotic environment; biotic and abiotic interactions; Concept of habitat and niche; niche width and overlap; fundamental and realized niche; resource partitioning; character displacement.

Ecosystem Ecology: Ecosystem structure; ecosystem function; energy flow and mineral cycling (C, N, P); primary production and decomposition; structure and function of some Indian ecosystems: terrestrial (forest, grassland) and aquatic (fresh water, marine, estuarine).

Species and Population Ecology: Characteristics of a population; population growth curves; population regulation; life history strategies (r and K selection); concept of metapopulation – demes and dispersal, interdemec extinctions; Types of species interactions, interspecific competition, herbivory, carnivory, pollination, symbiosis.

Community Ecology: Nature of communities; community structure and attributes; levels of species diversity and its measurement; edges and ecotones; Ecological succession – types; mechanisms; changes involved in succession; concept of climax.

Applied Ecology: Environmental pollution; global environmental change; biodiversity: status, monitoring and documentation; major drivers of biodiversity change (Climate change, greenhouse gases, carbon dioxide fertilization and sequestration); biodiversity management approaches; Concepts and utility of Ecological Niche Modeling, RS and GIS.

Unit-9: Tissue Culture and Genetic Engineering of Plants

Plant Tissue Culture and Organogenesis: Concept of cell differentiation and totipotency, single cell culture, techniques involved in and factors affecting single cell isolation; processes and factors affecting organogenesis, somatic embryogenesis.

Micropropagation and Somatic Hybridization: Concept and techniques of micropropagation–factors involved, applications and limitations; Protoplast isolation, culture and regeneration, somatic hybridization and selection; Utility of hybrids, cybrids and haploids.

Recombinant DNA Technology: Principles and techniques of gene cloning, restriction enzymes–characteristics and utility; Use of Vectors (plasmids, phages, phagemids and cosmids), artificial chromosomes (YAC); genomic and cDNA libraries, polymerase chain reaction (PCR) – principle, techniques and applications.

Genetic Engineering of Plants: Agrobacterium mediated and direct methods of gene transfer (electroporation and biolistic); applications of transgenic plants; biosafety – possible ecological risks and ethical concerns of GM crops; Genomics and proteomics–concept and applications, DNA microarrays, gene silencing mechanisms (RNAi), genome editing (CRISPR–Cas).

Unit-10: Biotechniques and Biostatistics

Microscopic Techniques: Visualization of cells and sub–cellular components by light microscopy; Microscopy of living cells. Principles and applications of Fluorescence and phase contrast microscopy; Scanning and Transmission Electron Microscopy.

Isolation, Detection and Sequencing of Macromolecules: Isolation, separation and analysis of RNA, DNA, Proteins, carbohydrate and lipid molecules; Gel Electrophoresis (AGE & PAGE and SDS–PAGE); DNA and protein sequencing methods, techniques and applications of RFLP, RAPD, AFLP, FISH and GISH.

Spectroscopy and Chromatography: Molecular analysis using UV/visible, fluorescence, NMR and ESR, Atomic Absorption and Atomic Emission spectroscopy; Chromatography techniques; Gas chromatography. Thin Layer Chromatography, HPLC and Mass Spectroscopy.

Biostatistics: Measures of central tendency and dispersion; Skewness, Kurtosis; methods of sampling (random and non–random); probability distributions (Binomial, Poisson and normal); Sampling distribution; Difference between parametric and nonparametric statistics; Confidence Interval; Errors; Levels of significance; Regression and Correlation; t–test; Analysis of variance; χ^2 test; Basic introduction to Multivariate statistics.

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